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$$CH_3CH_2COOAg + Br_2 \xrightarrow[\text{reflux}]{CCl_4} CH_3CH_2Br + CO_2 + AgBr$$

Silver propionate — Bromo ethane

14.3.4 Physical Properties

1. Pure haloalkanes are colourless. Bromo and iodo alkanes are coloured in the presence of light.

2. Haloalkanes having one, two or three carbon atoms are in the gaseous state at normal temperature. Haloalkanes having more than three carbon atoms are liquids or solids.

3). Boiling point and Melting point

i) Haloalkanes have higher boiling point and melting point than the parent alkanes having the same number of carbons because the intermolecular forces of attraction (dipole – dipole interaction and vander Waals forces) are stronger in haloalkane.

ii) The boiling point and melting point of haloalkanes decreases with respect to the helogen in the following order.

Example

$CH_3I > CH_3Br > CH_3Cl > CH_3F$

iii) The boiling points of chloro, bromo and iodo alkanes increase with the increase in the number of halogen atoms.

For Example:

$CCl_4 > CHCl_3 > CH_2Cl_2 > CH_3Cl$

iv) The boiling point and melting point of mono haloalkane increase with the increase in the number of carbon atoms.

Example

$CH_3CH_2CH_2Cl > CH_3CH_2Cl > CH_3Cl$

v) Among isomeric alkyl halides the boiling point decreases with the increase in branching in the alkyl group; with increase in branching, the molecule attains spherical shape with less surface area. As a result the inter molecular forces become weak, resulting in lower boiling points.

Example

$CH_3CH_2CH_2CH_2Cl > CH_3CH_2\underset{\displaystyle Cl}{\underset{|}{C}}HCH_3 >$

(351.5K) (341K)

$$CH_3-\overset{\displaystyle CH_3}{\overset{|}{\underset{\displaystyle Cl}{\underset{|}{C}}}}-CH_3$$

(324K)

4. Solubility

Haloalkanes are polar covalent compounds soluble in organic solvents, but insoluble in water because they cannot form hydrogen bonds with water molecules